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Microplastics transport in soils: A critical review

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ABSTRACT

Microplastics (MPs) in terrestrial environments are an emerging contaminant of high concern to ecosystems and human health. However, our understanding of the MPs' fate, particularly their transport within soils, remains elusive. This knowledge gap arises from the multiplicity of coupled physical, chemical and biological processes and parameters affecting MPs transport, together with the scarcity of systematic studies that aim to isolate their individual effects. In this paper, we provide a critical review of the state-of-the-art in our understanding of MPs transport, highlight knowledge gaps and suggest future research to bridge them. We classify the governing factors into four main categories: (i) MPs properties; (ii) soil physicochemical properties; (iii) hydrological conditions; and (iv) biological activity. Our analysis reveals that lack of clear trends in the dependence between MP transport and individual key parameters—often leading to contradictory findings—could be explained by the interference ("co-effects") with other parameters and processes.

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1. Introduction

Plastics are widely used globally due to their low manufacturing costs, ease of use, versatility, durability, and stability (Tábi et al., 2021). Annual production of plastics increased dramatically, from ~2 Mt. in 1950 to ~390 Mt. in 2021 (PEMRG, 2022). Plastics degrade into smaller particles through oxidation, hydrolysis, biodegradation, photodegradation, and mechanical processes (Sarkar et al., 2020). Microplastics (MPs), particles larger than 1 μm (Weber et al., 2022; >0.1 μm according to Koelmans et al., 2015) and smaller than 5 mm, can easily migrate by wind and water, making them extremely abundant in the atmosphere, soil, and water (Avio et al., 2017; Bullard et al., 2021), including in very remote areas (Ali, 2021). MPs are believed to pose a serious risk to living organisms including humans, animals, and plants (Leslie and Depledge, 2020; Zhang et al., 2022a; Djouina et al., 2023). Moreover, MPs can act as vectors to transport other contaminants adsorbed on their surface, such as heavy metals, pathogens (bacteria, fungi, protozoa, worms, viruses), and antibiotics (González-Pleiter et al., 2021; Liu et al., 2022; Lu et al., 2022; Wang et al., 2021), posing additional risks to the environment and human health.

Understanding MP transport in soil is important because of its broader implications for soil ecosystem and the environment. MPs have been shown to significantly affect soil structure (de Souza Machado et al., 2018), alter soil physiochemical properties (Medyńska-Juraszek and Jadhav, 2022), disrupt nutrient cycling (Rong et al., 2021), influence biodiversity (Omidoyin and Jho, 2023) and the bioavailability of heavy metals (Feng et al., 2022), and affect the transport of other pollutants such as antibiotics chlortetracycline hydrochloride and tetracycline (Xu et al., 2022b; Zhao et al., 2020). These effects have long-term implications for soil health and productivity due to MPs can persist in soil for decades.

MPs can be introduced into soils through various pathways. One prominent source is agricultural usage of plastic products such as plastic mulches, packaging materials, and plastic-based fertilizers (Fan et al., 2023). Other sources include leaching from waste landfills, direct (illegal) dumping of plastic materials, and irrigation and flooding of wastewaters, application of contaminated compost (Li et al., 2020b; Surendran et al., 2023). Additionally, the atmosphere contains airborne MPs from e.g. industrial activities and vehicle emissions, that can enter soils via precipitation and wind (Zhao et al., 2023).

There is a growing interest in the fate of MPs in both soil and aquatic environments (Dioses-Salinas et al., 2020; Wong et al., 2020; Fig. S1 in the Supplementary Information (SI)). A significant challenge in understanding the fate of MPs in soils is that, unlike in oceans, MPs can strongly affect the host (soil) properties—specifically the properties that control their transport (de Souza Machado et al., 2018; Guo et al., 2022; Wang et al., 2022a), which in turn affects the MP fate. Recent reviews focused on the interaction of MPs with contaminants in soils (Chang et al., 2022; Yang et al., 2022); the sources, distribution, behavior, and detection techniques of MPs in soils (Li et al., 2020b; Koutnik et al., 2021; Yan and Yang, 2023); transport and transformation of MPs and nanoplastics (NPs) in soils (Liu et al., 2021); the migration of MPs in the soil-groundwater environment (Ren et al., 2021); the effects of MP on soil properties (Wang et al., 2022a); the occurrence, fate, transport, and ecological risk of MPs in soils (Zhou et al., 2020; Huerta Lwanga et al., 2022; Li et al., 2020b; Xu et al., 2020); and the deleterious environmental effects of MPs in soils (Sajjad et al., 2022). Some reviews summarized the factors that affect MP transport in soil, e.g. including MP properties (e.g., size, shape, and type) and soil properties (e.g., porosity, texture, and composition) (Liu et al., 2021), as well as living organisms (Zhou et al., 2020). While these reviews shed light on how some of these properties affect MP transport, the impact of others remains elusive, with contradictory evidence of their effects that implies a more complex behavior of MPs than previously assumed. For example, Liu et al. (2021) indicated that smaller-size MPs have stronger mobility, in contrast to findings in subsequent studies (Qi et al., 2022; Gui et al., 2022). Recent research publications also pointed to further factors influencing MP transport, including hydrological conditions and living organisms (Gao et al., 2021; Xing et al., 2021; Gui et al., 2022; Li et al., 2021a; Luo et al., 2023b).

MP transport in soils is expected to follow the tortuous migration pathways associated with the soil microstructure. While these pathways could have both a vertical and horizontal component, many of the studies of MPs transport and the metrics used to describe it refer to vertical transport, possibly because of the increasing difficulty of soil remediation as MPs gets deeper (e.g. if it contaminates groundwaters (Helmberger et al., 2020)). An added challenge is the inconsistency in the MP transport/mobility metrics and the way they are measured (Zhang et al., 2022b), making comparison between studies challenging. MP mobility metrics can be categorized into (a) flux- and (b) depth-based, measured by either mass or number of particles (Table 1).

This paper provides a critical review of the state-of-the-art understanding of MP transport in soil. We point to several non-intuitive and often contradictory findings, demonstrating how these can be explained by the interference among factors and processes. Finally, we outline the knowledge gaps and propose future research directions.

2. Methodology and paper structure

To identify relevant papers, we conducted a literature search with the search term “microplastic” AND (“transport” OR “move” OR “mobile” OR “mobility”) AND (“soil” OR “porous media” OR “sand”) – applied to title, abstract, keywords and full publication text. The search was conducted using the Google Scholar, PubMed, and Web of Science databases, and focused on papers published since 2000 (search last updated 22 July 2024; see Table S1 in SI for search details). Results were sorted according to relevance, and the searched literature was manually screened for papers that explicitly investigated the factors that affect microplastic transport in soil, excluding papers that coincidentally met the search criteria while being nonrelevant (e.g. papers that focused on NPs but mention “microplastics” in the text). Similarly, papers that focused on MP transport in aquatic environments (while mentioning briefly soils) or on the effect of MPs on transport of other substances were also excluded. Overall, we retained 113 papers that specifically addressed MP migration in soil, including 50 experimental studies. We note that our search results include papers that define particles in the 0.1 μm – 1.0 μm range as “microplastics” (as per the definition of Koelmans et al., 2015), while excluding papers that define such particles as “nanoplastics” (as per the definition of Weber et al., 2022).

Many of the studies on MP transport in soil were conducted using artificially constructed soil analogs such as quartz sand or glass beads. These experiments, by simplifying visualization and/or analysis,

Table 1
MP mobility metrics used in the reviewed literature.

	Metric name	Definition	Unit	References
Flux-based	Transport rate / Mass recovery / Mass balance / Total mass recovery/ penetration rate	weight of MPs traversing sample [a], relative to the initial MP weight	%	Cao et al., 2023; Cai et al., 2019; Chu et al., 2019; Dong et al., 2021, 2022; Fei et al., 2022; Gui et al., 2022; Hou et al., 2020; He et al., 2020, 2021; Jiang et al., 2021, 2022; Li et al., 2019, 2023; Li et al., 2020a; Li et al., 2021b; Liu et al., 2024; Rong et al., 2022, 2023; Sun et al., 2022; Tong et al., 2020; Wu et al., 2020; Wang et al., 2022b; Wang et al., 2022d; Wang et al., 2022f; Wang et al., 2022e; Xu et al., 2022a, 2022b; Xu et al., 2024a; Xu et al., 2024b; Zhao et al., 2020; Zhao et al., 2022a; Zhou et al., 2024
	Loss ratio	number of mobilized MPs, relative to initial number of MPs	%	Rehm et al., 2021
	Maximum depth ratio	number of MPs at maximum depth reached, relative to the initial number of MPs	%	Cohen and Radian, 2022
	Relative mobility	number of MPs recovered at a given depth, relative to total number of mobilized MPs (soil + solution)	%	Zhang et al., 2022b
	Concentration in soil	concentration of MPs (in weight, or number) per soil mass, at a given soil depth	mg/g (or items/g)	Xing et al., 2021; Zhao et al., 2022b
	Concentration in solution	concentration of MPs (number per volume) [a]	items/mL	Qi et al., 2022
	Normalized concentration	concentration of MPs [a], relative to initial MP concentration	–	Ivanic et al., 2023; Koutnik et al., 2022b; Wang et al., 2022c;
Depth-based	Maximum depth	maximum depth reached by mobilized MP particles	cm	Gao et al., 2021; Mancini et al., 2023; Gunther et al., 2023; O'Connor et al., 2019; Ranjan et al., 2023; Tumwet et al., 2022; Wang et al., 2022b; Waldschläger and Schüttrumpf, 2020; Yan et al., 2020; Zhang et al., 2022b; Zhao et al., 2022b

(a) measured in the solution collected at the sample outlet.

provide invaluable insights into the transport of MPs in natural soils and thus are included in this review and analysis. However, contributions on controlled systems with even more idealized media (such as microfluidic channels), aimed to enhance fundamental understanding of specific aspects (e.g., MP entrainment in liquid films; Wu et al., 2021), did not fall within our search criteria and hence not included in this review. In the rest of the paper, when making general statements for which the distinction between natural soils and artificial soil analogs is not crucial, we use the terms soil and porous matrix interchangeably. We also stress that in some experiments, the settings used could be different than in typical natural conditions, for instance in terms of how the water and MPs were introduced, or the MPs concentration. Our review includes these papers since some of the parameters in the natural environment, for instance, rainfall intensity, soil solution geochemistry, soil pore and particle sizes, MPs sizes, and concentration can fall within a huge range of values, making the insights from these studies invaluable. Even for parameters for which there is a relatively narrow range which considered typical, with time these ranges could be substantially extended; as an example, while the initial MP concentration is typically considered to be of only a few %, their transport in heterogenous soils containing preferential pathways could create "MP hotspots" with much higher MP concentrations (Cramer et al., 2023). The paper is structured such that each section focuses on the impact of a group of properties on MP transport while pointing to their coupling with others. Sections 3–5 overview the impact of MP properties, soil properties, and the hydrological conditions, respectively. Section 6 overviews the effect of living organisms. Section 7 provides concluding remarks, summarizing the research gaps and suggesting future directions. Additionally, Tables S2 and S3 in the SI list the experimental conditions classified in terms of the MP and soil properties, respectively.

3. Microplastic properties

This section focuses on the following MP properties: size, type, and shape, and their derivatives such as density, hydrophobicity, surface roughness, hydrodynamic diameter, and Zeta potential. We also discuss here MP aging, as a process altering other MP properties.

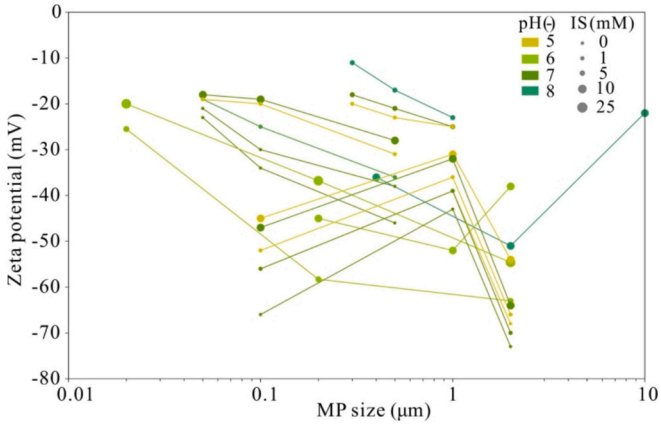


Fig. 1. Zeta potential vs. MP size under different pH and ionic strength (IS) conditions. Data from Li et al. (2019), Wang et al. (2022c, 2022e), Rong et al. (2022), Gui et al. (2022), Cai et al. (2019), cf. Table S4 (SI).

3.1. Microplastic particle size

A basic property that has a substantial influence on MP mobility in soil is MP size. Since MP size affects multiple mechanisms which control MP mobility, there is a complex trend: in general, MP mobility in soil for larger MPs (≳1 μm) is inversely proportional to the MP size, and it was found in terms of different MP mobility metrics (Table 1), including maximum depth (Ranjan et al., 2023; Gao et al., 2021; Zhang et al., 2022b; O'Connor et al., 2019; Tumwet et al., 2022), proportion of MPs reaching the maximum depth (Cohen and Radian, 2022), mass recovery rate (Rong et al., 2022), and normalized concentration (Wang et al., 2022c). This is rather intuitive: for a given size of soil pores, smaller MPs can traverse further (Gao et al., 2021; Ranjan et al., 2023; O'Connor et al., 2019). In contrast, Qi et al. (2022) found that 20 μm MPs had higher mobility than 10 μm MPs, i.e. when water flow rate was 4 mL/min, the MP concentrations of 10 and 20 μm MPs, went through soil, were 0.85 and 1.49 items/mL, respectively. The authors explained that

smaller MP particles have higher affinity to soil particles, which reduces MP mobility. For 0.1 μm ≲ MP ≲1 μm, increasing MP size increases MP mobility (in terms of transport rate in Gui et al. (2022)). The reason that the trend in MP size vs. MP mobility reverses for very small MPs is linked to the fact that for very small particles, other forces and mechanisms become dominant. In particular, the Zeta potential, a measure of the electrical charge at the surface of particles suspended in solution (Bhattacharjee, 2016), plays an important role. For example, in several studies, for MPs ≲1 μm, the Zeta potential of MP becomes less negative with decreasing MP size (Fig. 1; Wang et al., 2022e; Gui et al., 2022), and, therefore, decreasing the repulsive force between MPs and soil particles, enhancing their retention in soil (Gui et al., 2022). However, the experiments from Cai et al. (2019) show a different result whereby the Zeta potential of 0.2 μm MP is more negative than that of 1 μm (Fig. 1). The relationship between MP size and MP Zeta potential is also related to environmental factors such as pH and ionic strength of soil solution (Fig. 1; Table S4 in SI).

At even smaller plastic particle sizes (<0.1 μm), different mechanisms can become dominant, thus reversing the trend again. For plastic particles smaller than 0.1 μm, Wang et al., 2022e and Li et al. (2019) showed that mobility increased with decreasing particle size (smaller particles exhibited higher total mass recovery and mass recovery rate). The authors attributed this to hydrodynamic diameter, which is the diameter of the particles including their electronic layers (Maguire et al., 2018), implying this mechanism can become dominant for very small particles. The dependence of MP mobility with MP size is intimately linked with the MP concentration and soil pore sizes; see discussion in Sections 3.4 and 4.1. As a result of the aforementioned interdependencies among mechanisms, establishing the critical size at which the MP mobility trend reverses is not straightforward, as it can depend on other various properties. This can explain the apparent discrepancy between a critical size of 2 μm (for which MP normalized concentrations were higher than for 0.4 μm and 10 μm; Wang et al., 2022c) and 1 μm (for which MP mass recovery was greater than for 0.2 μm and 2 μm; Rong et al., 2022).

3.2. Microplastic shape

Microplastics are found in natural soils in a variety of shapes (Fig. S2 in SI), with the most common being fragments (*irregular*-shaped), fibers, films, foams, and beads (spheres) (Sajjad et al., 2022). While it is accepted that MP shapes affect their transport, the link between MP shapes and their transport in soils remains elusive, also due to the large number of additional parameters which vary among experiments probing the impact of MP shapes (Tables S2-S3 in SI). Beads were found to reach deeper than films because films were trapped more easily by plant roots (Li et al., 2021a). Fragments were found to move deeper than fibers (Gao et al., 2021; Cohen and Radian, 2022; Waldschläger and Schüttrumpf, 2020), possibly due to the tendency of fibers to become entangled with soil grains (de Souza Machado et al., 2018; Waldschläger and Schüttrumpf, 2020; Keller et al., 2019). Similarly, spheres reach a deeper layer than fragments (irregular-shaped) because fragments can easily be entangled with soil particles (Waldschläger and Schüttrumpf, 2020). Zhang et al. (2022b), fibers reached deeper than fragments, which in turn were more mobile than foams and films. However, it is hard to draw conclusive evidence regarding MP fiber mobility from Zhang et al. (2022b), as the authors did not specify the fiber diameter, which has a greater effect on MP infiltration than its length (Waldschläger and Schüttrumpf, 2020). The above suggests that fiber, fragments, and beads tend to migrate deeper than films and foams.

3.3. Microplastics type

Different MP types (made from different polymers) have different properties including density, hydrophobicity (contact angle), surface roughness, and Zeta potential (O'Connor et al., 2019; Fei et al., 2022; Ranjan et al., 2023; Gao et al., 2021; Cohen and Radian, 2022). Each of these properties can affect various aspects of MP transport. In general, heavier MPs have greater vertical mobility. For example, polyethylene (PE; density of 0.884–0.898 g/cm^3) reached deeper than polypropylene (PP; density of 0.833 g/cm^3) (O'Connor et al., 2019). However, surface properties often dominate over density: lower hydrophobicity, higher Zeta potential, and lower roughness were found to have a stronger positive effect on MP transport in soil than higher density (Fei et al., 2022; Ranjan et al., 2023; Gao et al., 2021; Cohen and Radian, 2022). Despite its lower density, the mass recovery rate of polylactic acid (PLA; density of 1.25 g/cm^3) was higher than polyvinyl chloride (PVC; density of 1.38 g/cm^3), because PLA has higher Zeta potential and lower hydrophobicity (Fei et al., 2022). Comparing PE, polyethylene terephthalate (PET), and PP, the order of PE > PET>PP was identified in terms of depth reached (Ranjan et al., 2023; Gao et al., 2021), again explained by the lower hydrophobicity and higher Zeta potential of PE vs. PET vs. PP, despite its lower density (all other conditions were uniform). The lower mobility of PP was explained by its low density and Zeta potential, and its higher hydrophobicity and surface roughness (Ranjan et al., 2023; Gao et al., 2021). Lower hydrophobicity was used to explain deeper penetration of polyamide (PA) (vs. PE, PET, and PP in Gao et al., 2021; and vs. PET in terms of depth ratio in Cohen and Radian, 2022). Inconsistencies between studies, e.g. deeper penetration of PET vs. PE in Gao et al. (2021), opposite to the trend found by Zhang et al. (2022b), could be due to the different mobility metrics: PET exhibited larger penetration depth than PE, however its relative mobility was lower (Zhang et al. (2022b)). Additional potential sources of inconsistency are due to using different analytical methods for sample preparation and analysis of MPs in the MP transport experiments (Table S5). Differences in other experimental parameters, including the soil properties, hydraulic conditions, and MP size and shape might also cause inconsistency between studies.

3.4. Microplastic concentration

The concentration of MPs does not always correlate with their mobility. Xu et al. (2022a, 2022b) showed that higher concentration of MPs (for two sizes, 0.1 μm and 2 μm) resulted in a higher mass recovery rate. This is due to the adsorption of some MPs on soil particles which then depleted the available sites for adsorption with increasing MP concentration and, therefore, enhancing MP mobility. Contrasting findings in smaller particles (≲ 0.1 μm), where the mobility (total recovery rate) showed inverse correlation with their concentration (Wang et al., 2022e). Contradictory findings for plastic particles ≲0.1 μm with increasing their concentration are probably due to the increasing dominance of mechanisms as plastic particle size decreases, including: (i) aggregation and deposition (Wang et al., 2022e); (ii) adsorption on soil particles due to the increase in plastic particle specific surface area (Shen et al., 2019); as well as (iii) clogging of pores by plastic particles. This contraction may also be due to differences in other MPs and soil properties between these studies (Tables S2-S3 in SI). Flow rates and soil particle size can also interplay with MP concentrations to affect MP transport. For example, in 1–2 mm sand, while at high-enough water flow rates (2.5 mL/min in Hou et al., 2020) MP mobility shows the expected proportionality with MP concentration, at lower flow rates (1–2 mL/min) no such correlation was found (Hou et al., 2020). A possible explanation is that MPs can be more easily intercepted by sand particles at slow flow rates, with faster flow rates allowing easier flushing of MPs reducing such interceptions (Hou et al., 2020). Additionally, at a water flow rate of 1–1.5 mL/min, the MP normalized concentration increased with increase of MP concentration in 2–4 mm sand, while at 1–2 mm sand no such correlation was found. A possible explanation is that the blockage of MP particles in finer soils is more apparent (Hou et al., 2020).

3.5. Microplastic aging

Aging of MPs, which can occur by oxidation, hydrolysis, biodegradation, photodegradation, and mechanical processes, can alter a variety of physical and chemical properties of MPs. The mobility of aged MPs was found to be greater than that of pristine MPs, in terms of both MP concentration (Qi et al., 2022) and maximum depth (Wang et al., 2022b). One reason is that aging increases MP oxygen-containing groups (Ali et al., 2023), which can reduce MP adsorption on soil particles (Qi et al., 2022; Yan et al., 2020). Furthermore, aging alters polymer structure, reducing MP size and changing their shape (Ren et al., 2021; Yan et al., 2020). For example, Mao et al. (2020) found that the C—H bond of microplastics breaks under the action of UV light, causing MP to break into smaller particles (Mao et al., 2020). Breaking into smaller MPs, which is known to enhance MP mobility (Ranjan et al., 2023), as well as reduce MP hydrodynamic diameter (Wang et al., 2022b). Aging also increases MP Zeta potential, and hence the interaction energy between MPs and soil particles (Wang et al., 2022b), and can increase MP surface hydrophilicity, promoting MP transport in soil (Gunther et al., 2023; Li et al., 2021a). Notably, aging can increase the surface roughness of MP (Luo et al., 2023a), which may lead to interlocking between the rough surfaces of MPs and the irregularly shaped soil particles, thereby decreasing MP mobility. Further research is needed to evaluate the relative impact of this mechanism vs. that of other, competing effects of MP aging (which facilitate MP migration) including reduced size, increased surface functionality, and enhanced hydrophilicity.

4. Soil physicochemical properties

This section reviews the impact of soil physical and chemical properties including soil texture, grain/pore size, soil pH, ionic strength (IS), soil mineralogy, and organic components, on MP transport. We also include here the impact of chemical substances present in the soil, as, while they are not soil properties per se, they do affect soil chemistry.

4.1. Soil texture, grain and pore size

Overall, coarser soil texture promotes MP transport; this was found in terms of mass recovery rate (Dong et al., 2022; Wu et al., 2020), maximum depth (Mancini et al., 2023), transport rate (Gui et al., 2022), MP concentration (number of MPs per kg soil) at maximum depth (Xing et al., 2021), and loss ratio (Rehm et al., 2021). For a given MP size, larger soil particles, which in turn imply larger pore sizes, enhance MP mobility in soil, in terms of mass recovery rate (Dong et al., 2022; Hou et al., 2020), maximum depth (Gao et al., 2021; Ranjan et al., 2023), and MP concentration (Qi et al., 2022). The control parameter when varying both MP size and soil texture can be evaluated using the ratio of the diameter of MP particles (D_{MP}) vs. that of the soil particles (D_S), where a smaller ratio promotes MP transport in soil (Waldschläger and Schüttrumpf, 2020). The critical ratio can be evaluated theoretically by considering the size of a constriction in the simple case of a hexagonal packing of equal-sized spheres: the narrowest aperture (between each triplet of particles) is $\sim 0.1D_S$, indicating a critical ratio of ~0.1. Experimental data agrees with this estimation: MPs were found to be highly mobile for $D_{MP}/D_S < 0.1$, and nearly immobile for $D_{MP}/D_S > 0.3$ (Gao et al., 2021; Ranjan et al., 2023). Similarly, the infiltration depth of MPs was found to reduce from 13.1 cm to 4.63 cm, and 1.18 cm at $D_{MP}/D_S < 0.11$, $0.11 < D_{MP}/D_S < 0.32$, and $D_{MP}/D_S > 0.32$, respectively. However, we note that because most soils are composed of a wide range of particle sizes, the average grain diameter D_S is insufficient to describe the impact of soil texture on MP mobility, which may require to consider a more comprehensive parameter such as particle size distribution. In addition to the obvious effect of larger soil pores in coarse-textured soils (Dong et al., 2022; Xing et al., 2021), larger soil particles have lower specific surface area, and therefore, adsorb and retain MPs less than finer soils. Soil texture also affects soil Zeta potential. In most cases, both soil and MP surfaces are negatively charged; in such cases, coarser soils have higher Zeta potential (Dong et al., 2022; Rong et al., 2022; Gui et al., 2022; Li et al., 2023), and thus stronger repulsion promoting MP mobility (in terms of mass recovery rate in Wu et al., 2020). MP surface roughness was found to reduce MP penetration depth, due to increased friction with the soil pores (Ranjan et al., 2023). However, this effect is subject to interference with other mechanisms and properties, including water flow rate and MP concentration (Hou et al., 2020). Higher porosity was also found to promote MP mass recovery rate (Li et al., 2023; Dong et al., 2022). We stress however that porosity, being a scalar which represents a sample-averaged property, is a too simplistic soil characteristic to provide a clear trend of MP mobility. For instance, soil can have high porosity which is poorly connected (e.g. with dead-end pores), such that it has limited ability to transport water and MPs. This could potentially explain the seemingly-contradictory result of higher MP mass recovery rate in low-porosity desert soil than in high-porosity red soil (Wu et al., 2020).

4.2. Mineralogy of soil grains

Soil mineralogy affects MP transport in two ways. Firstly, forming MP-minerals aggregates, which have a higher density than MPs alone, can enhance the depth of MPs penetration (Yan et al., 2020). Secondly, MP-minerals aggregates can decrease MP mobility in soil. For instance, Li et al. (2019), Li et al. (2020a), and Gui et al. (2022) found that the forming of MP-Fe/Al oxide aggregates decreased the MP mass recovery rate and transport rate because the diameter of the MP-Fe/Al oxide aggregates was larger than individual MP particles. The effect of MP-minerals aggregates on MP transport might be MP size dependent (< 5 mm in Yan et al., 2020, vs. 0.02–2 μm in Li et al., 2019; Li et al., 2020a, vs. 0.3–1 μm in Gui et al., 2022). Koutnik et al. (2022a) found that the MP density plays a greater role for the movement of larger MPs (> 50 μm); however, the intermolecular force plays a greater role than MP density for the movement of small MPs (<50 μm).

Soil minerals affect MP transport by affecting the MP Zeta potential and the interaction of MPs and minerals (Yan et al., 2020; Li et al., 2019; Li et al., 2020a; Li et al., 2021b; Wu et al., 2020; Gui et al., 2022). Yan et al. (2020) found that natural soil minerals (soil minerals were not specified by authors) enhanced the penetration depth of MPs because they make the MP Zeta potential more negative. In contrast, Wu et al. (2020) and Gui et al. (2022) found that the presence of Fe or Al oxide minerals (Fe/Al oxides) decreased MP mobility (both mass recovery rate and transport rate) because MPs were adsorbed on the surface of Fe/Al oxides. Fe/Al oxides were also found to limit MP transport by reducing the volume of larger pores (Lu et al., 2014). Thus, highly weathered soils such as Ultisols, which typically contain significant quantities of Fe and Al oxides, may act as barriers to MP movement. The contrasting findings of Yan et al. (2020) versus Wu et al. (2020) and Gui et al. (2022) might be due to difference in the electrical properties of different minerals, as well as the different mobility metrics (rate- vs. depth-based). The effect of Fe and Al oxides on MP transport is pH-dependent because their surface charge is altered depends on pH: it is positive for acidic and neutral pH, and it becomes negative above pH 8 (for example in hematite, due to the deprotonation of Fe-OH; Nie et al., 2023). Kaolinite and bentonite particles were found to increase the mass recovery rate of MPs at pH 6 and IS 5 mM NaCl because they can act as a carrier for MPs, thus promoting MP mobility (Li et al., 2020a; Li et al., 2021b). However, bentonite would decrease the MP mobility at IS of 25 mM NaCl because the diameter of MP-bentonite aggregates was around 3 times larger than individual MP particle (Li et al., 2021b).

4.3. Organic components in soils

Soil organic matter enhances MP transport (Ivanic et al., 2023; Gao et al., 2021; Dong et al., 2021; Hou et al., 2020; Wang et al., 2022e; Xu et al., 2022a, 2022b; Zhao et al., 2022a; Xu et al., 2024b). For instance,

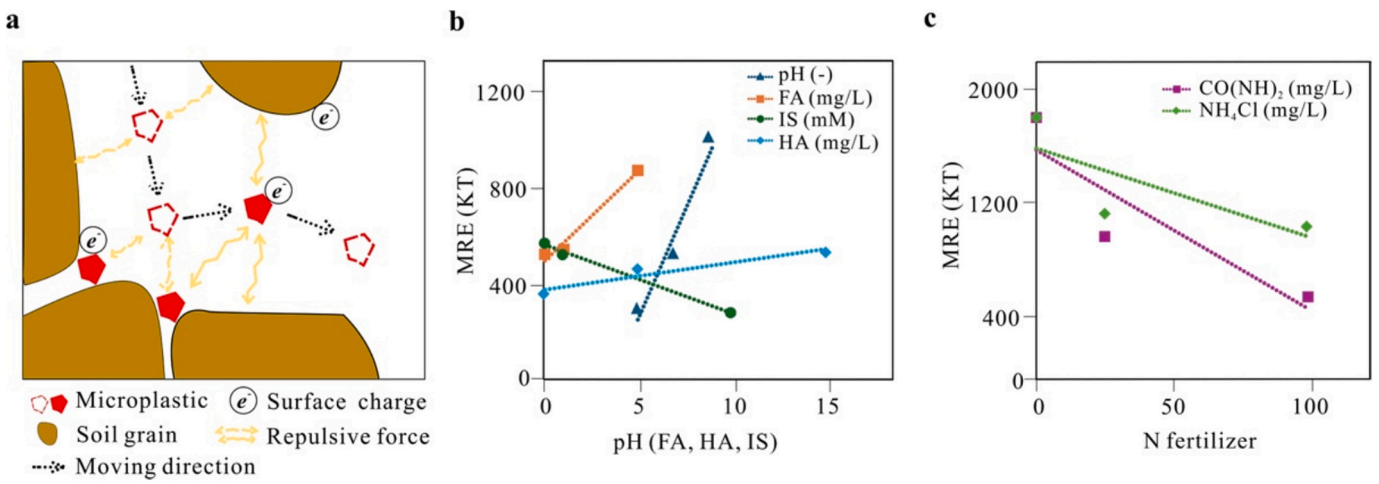


Fig. 2. Schematic diagram of the interaction force between MPs and soil particles (a) and the relationship between maximum repulsive energy (MRE), pH, fulvic acid (FA), humic acid (HA), ionic strength (IS)(b), $CO(NH)_2$, and NH_4Cl(c). Data in (b and c) is from Wang et al. (2022e), Wang et al. (2022f), and Rong et al. (2022).

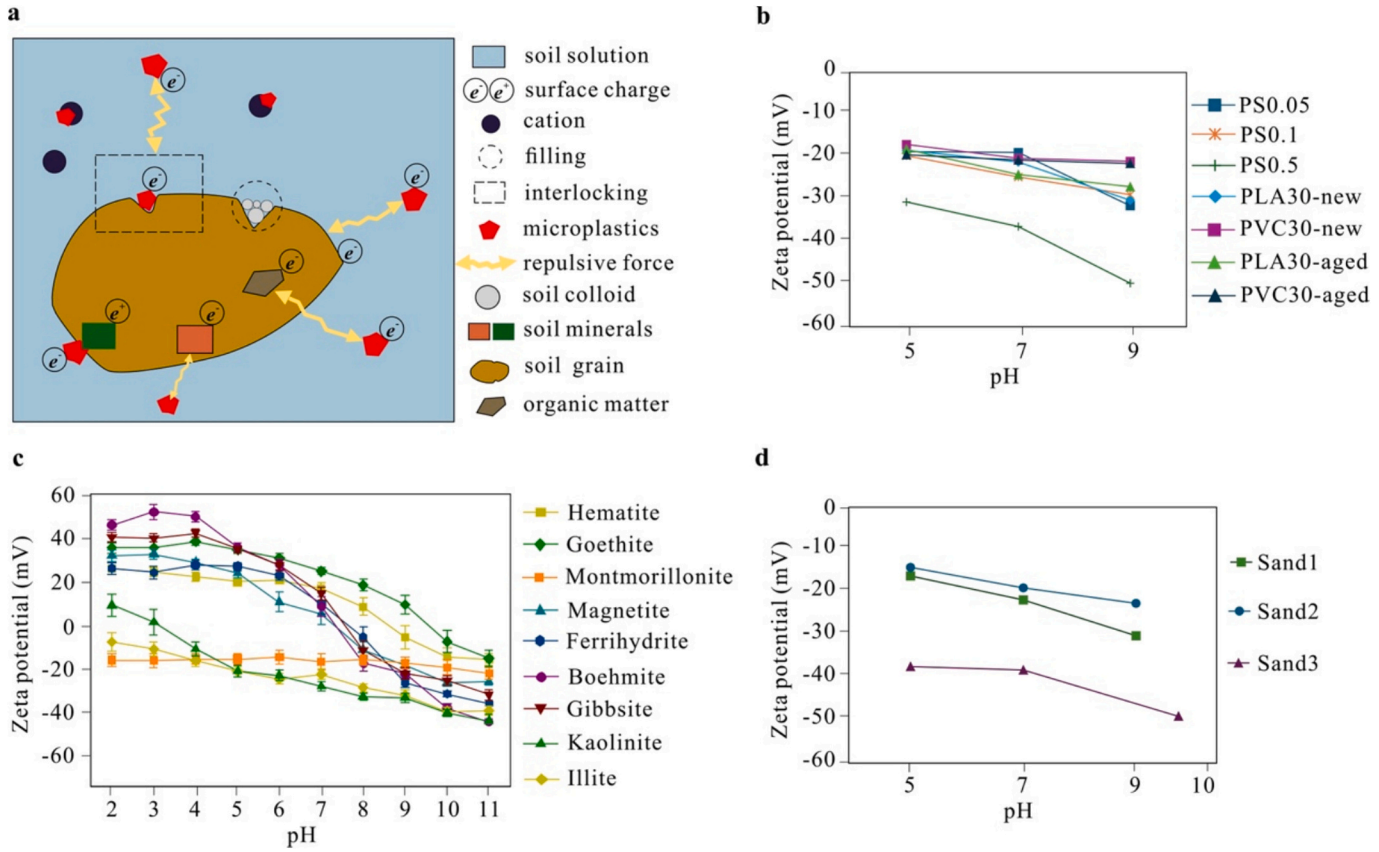


Fig. 3. Conceptual diagram of the interaction of MPs with soil solution (a), and the relationships between pH and MP Zeta potential, for various MPs (type-size-age; b), soil minerals (c), and sands (d). Data in (b,d) from Wang et al. (2022e), Fei et al., 2022; and Wu et al., 2020; and in (c) from Nie et al., 2023.

dissolved organic matter was found to increase MP normalized concentration by increasing the MP surface wettability and increasing the dispersion of MP particles on the soil pore surfaces (Ivanic et al., 2023; Xu et al., 2024b). Humic acid (HA) was found to enhance the maximal MP penetration depth by increasing repulsion between the MPs and soil particles and possibly decreasing MP hydrophobicity (Fig. 2b; Gao et al., 2021). Further studies showed that HA decreased the roughness of the MP surface, promoting the MP mass recovery rate (Zhao et al., 2022a; Dong et al., 2021, 2022). Similarly, fulvic acid (FA) was found to increase MP normalized concentration by increasing Zeta potential of both MPs and sand particles and thus their repulsive force (represented in terms of maximum repulsive energy controlling adsorption; Wang et al., 2022e; Fig. 2a~b). However, Hou et al. (2020) demonstrated that the MP mass recovery rate was not always increased with increasing FA concentration because for high FA concentration (> 5 mg/L) other factors such as hydrophobicity and buoyancy can promote MP transport over electrostatic forces (Hou et al., 2020). HA has greater transport-promoting for MPs than FA because HA has a larger molecular weight and more complex structure, leading to a stronger repulsive force to MP particles than that of FA (Liu et al., 2024). Soil colloids were shown to

promote MP mass recovery rate by filling the concave area of the soil pores (in the contact between the grains), reducing the soil surface roughness, and increasing the repulsive force between MPs and pore walls (Fig. 3a; Xu et al., 2022a, 2022b). Soils rich in organic matter are conducive to MP transport: beyond the direct interaction between MP and organic matter, the organic matter also improves soil structure by forming secondary aggregates and enhancing soil biological activity (Tarafdar, 2022), creating stable porous pathways for MP transport. Notably, commonly used agricultural organic supplements (e.g. manure and sewage sludge) also contain MPs (Zhang et al., 2022c).

4.4. pH

Both MP mass recovery rate and normalized concentration were found to increase with increasing pH (Fei et al., 2022; Wu et al., 2020; Wang et al., 2022e). This is due to the Zeta potential of both MPs and soil minerals becoming more negative with increasing pH (Fig. 3b∼d), indicating a stronger repulsive force between them (Fig. 2a∼b). The repulsion (associated with electrostatic forces) was suggested as the dominant factor promoting MP mass recovery rate with increasing pH (Wu et al., 2020). Fei et al. (2022) and Wang et al. (2022e) found that the MP hydrodynamic diameter increases with decreasing pH and, therefore, decreasing MP mass recovery. Nie et al. (2023) found that increasing pH from 3 to 8.5 reversed the interaction force of MPs and minerals (from attraction to repulsion), decreasing MP-minerals aggregation (Nie et al., 2023). Overall, the repulsive force between MP and sand as well as other soil minerals increases with increasing pH (Fig. 2b).

4.5. Ionic strength

In general, MP mobility decreases with increasing IS (Chu et al., 2019; Dong et al., 2022; Hou et al., 2020; Wu et al., 2020; Wang et al., 2022e; Li et al., 2021b; found in terms of both MP mass recovery rate, and MP normalized concentration). The Zeta potential of both soil particles and MPs increases with increasing IS (Wang et al., 2022e; Hou et al., 2020; Cao et al., 2023). The repulsive force between sand and microplastic increased with decreasing IS (Chu et al., 2019). Moreover, the hydrodynamic diameter of MPs increases with increasing IS, which leads to enhanced retention and reduced transport of MPs (Cao et al., 2023). Different types of MPs can show relatively different mobility at different IS due to specific characteristics (e.g., chemical composition; presence of functional groups on the surface of MPs). For example, Cao et al. (2023) found that the total mass recovery of PVC and PP MPs was relatively lower than PMMA and PET MPs for increasing IS. It was found that Ca^{2+} has a greater effect on MP retention than K^{+}, Na^{+}, and Mg^{2+} (Dong et al., 2021), and that Al^{3+} has a greater effect on MP retention than Na^{+} and Mg^{2+} (Fei et al., 2022) because cations with higher ionic radii lead to stronger charge screening, which increases particle retention (Dong et al., 2021; Wu et al., 2020). In soil contaminated by MPs, different IS have different effects on the MP migration because the degree of aggregation of soil mineral particles with MPs can be different under different IS. For example, ∼77 % and ∼ 97 % of bentonites were adsorbed onto MPs forming MP-bentonite hetero-aggregates when IS was 5 mM and 25 mM NaCl, respectively; the diameter of the MP-bentonite hetero-aggregates increased with increasing IS, which in turn decreased the MP mass recovery rate (Li et al., 2021b).

4.6. Chemical substances in soils

Chemical substances that might be present in soil, such as nitrogen fertilizers, antibiotics, nonylphenol, emerging pollutants (e.g., perfluorooctanoic acid (PFOA), as part of *per-* and polyfluorinated substances (PFAS)), and biochar, can also impact MP transport. Nitrogen fertilizers were found to decrease the MP mass recovery rate because they decreased the Zeta potential of both MPs and soil particles and increased the hydrodynamic diameter of MPs (Fig. 2a, c; Rong et al., 2022). Chlortetracycline hydrochloride (a widely used antibiotic in agriculture) was shown to decrease the MP mass recovery rate, due to a decline in the electrostatic repulsion between MP and sand grains (Xu et al., 2022b). The large specific surface area of biochar was shown to provide more adsorption sites for MPs, hindering their transport (Wang et al., 2022d), with the stronger impact of Fe_3O_4 biochar because its surface roughness, specific surface area, and the total pore volume are much higher than normal biochar, indicating that the Fe_3O_4 biochar is more beneficial to the MP retention (Wang et al., 2022f; Tong et al., 2020). Unlike biochar and nitrogen fertilizers, nonylphenol (a non-ionic surfactant present in pesticide formulation) increases the MP mass recovery rate, as it increases the MP Zeta potential, and possibly because of the hydrogen bonds forming between MPs and nonylphenol (Xu et al., 2022a).

The presence of surfactants in soil alters the surface hydrophobicity of microplastic, influencing MP transport (Sun et al., 2022; Jiang et al., 2021). Both cationic surfactants and anionic surfactants were found to promote the MP mass recovery rate, as long as the surfactant concentration remained lower than the critical micelle concentration; this was explained by the increase in the Zeta potential of MPs and sand particles and the decrease of MP hydrophobicity by the surfactants (Jiang et al., 2021). Cationic surfactants were found to have a greater effect on MP transport than anionic surfactants because the former provides more osmotic and elastic repulsion forces between MPs and sand particles (Jiang et al., 2022). Surfactants have a nontrivial effect with MP types (PE and PP) on MP mobility: while the mobility of PE was higher than of PP without surfactants (in terms of maximum depth; Gao et al., 2021; O'Connor et al., 2019; Ranjan et al., 2023), the addition of surfactants reversed that trend (in terms of MP mass recovery rate; Jiang et al., 2021, 2022; Zhou et al., 2024). The authors explained this by the increase in Zeta potential, as well as the larger surface area and roughness of PP MPs, promoting surfactant retention (Jiang et al., 2021). Zhou et al. (2024) attribute this to the PP hydrophilicity being higher than PE hydrophilicity when adding surfactants. However, we note that the use of two different MP mobility metrics across these studies does not allow a decisive conclusion regarding the main reason for the reversal in mobility of PP and PE. Zhao et al. (2020) found that the effect of tetracycline (TC) on the of MP mobility depends on the type of solution and IS. For example, in KCl or $CaCl_2$ solutions at IS 1 mM, TC increased the mass recovery of MP because TC decreased MP hydrophobicity, promoting MP mobile. However, at IS 100 mM, TC decreased the MP mobility in the KCl solution but increased the MP mobility in the $CaCl_2$ solution. The authors attributed this to the presence of TC rendered the attachment efficiencies of MP less sensitive to the changes in the $CaCl_2$ solutions (Zhao et al., 2020). Similarly, titanium dioxide nanoparticles (nTiO$_2$) decreased the mass recovery of MP because they formed aggregates with MPs, which increased the diameter of MP compared to individual MP (Cai et al., 2019). PFOA (a widely used industrial surfactant) was found to decrease the normalized concentration of negatively-charged MPs, thereby decreasing their Zeta potential, and promoting their adsorption on soil particles; an opposite effect—increasing normalized concentration—was observed for positively-charged MPs (carboxyl group modified MPs), due to the reversal of their surface charge, limiting their adsorption on soil particles (Rong et al., 2023). In contrast, Xu et al. (2024a) found that PFOA increases the recovery rate of MPs because PFOA induces a more negatively charged MP surface.

5. Hydrological conditions

Since transport of MPs in soils is driven by water infiltration, the hydrological conditions that control how water is introduced into the soil play a decisive role. The interaction of MPs with water is complex: in addition to the amount of water and its velocity, the transport of small MPs also depends on the motion of air-water interfaces in the vadose zone. The fact that MPs are hydrophobic further complicates their

migration, as explained below.

5.1. Water flux

Higher water flux in general promotes MP transport (Fei et al., 2022; Qi et al., 2022; Wang et al., 2022c; Dong et al., 2022). This is mainly due to the increase in the shear force which sweeps MPs with the water flow; however, slower water infiltration allows more transverse (horizontal) flow, which could enhance MP transport by allowing the water to traverse larger MPs-contaminated areas in the soil (shown in terms of mass recovery rate in Hou et al. (2020) and Fei et al. (2022)). Higher water velocity was shown to increase the detachment of MPs from the soil matrix (Wang et al., 2022c; Qi et al., 2022; Dong et al., 2022). Further intricacies can arise due to the interactions of flow rates with factors including IS, MP concentration, and pore size (Dong et al., 2022; Hou et al., 2020). Dong et al. (2022) showed a positive correlation between water velocity and MP mass recovery rate under high IS (10 Mm NaCl), albeit with negligible influence of velocity on MP transport at lower IS (0.1 Mm NaCl).

Another nontrivial relationship between MP transport and water flux is linked to MP concentration and soil particle size, showing an initial increase in MP mass recovery rate with velocity followed by a decrease at even higher velocities, for MP concentrations of 0.3 mg/L and 0.5 mg/L in 1–2 and 2–4 mm sand, respectively (Hou et al., 2020). The authors attributed this decline in MP mobility to an increase in MP adsorption to soil particles due to increase in frequency of collisions between MP particles with flow rate, occupying a larger portion of the pore space. All these factors increase the probability of MPs to adsorb on the soil particles. We note most studies on the effect of flow rate were conducted in coarser soils, where flow rates would be higher than in finer soils.

5.2. Soil water saturation

Soil water saturation (the relative pore volume occupied by water, changing by wetting/drying) also affects MP transport (Dong et al., 2022). This might be due to the hydrophobic nature of MPs, minimized contact of MPs with water, and MPs can remain trapped within the air-water interface (Al Harraq and Bharti, 2022). As decrease in water saturation produces more air-water interfaces, this in turn limits MP migration in soils (Dong et al., 2022). Another reason for this trend is the increase in trapped air bubbles, that decrease the available pore space for migration of water and MPs (Dong et al., 2022). For low density MPs, however, increasing water saturation was shown to decrease MP mass recovery rate because of buoyancy effects (O'Connor et al., 2019; Li et al., 2021a).

5.3. Wetting-drying cycles

Since MP transport is strongly linked to water saturation (Dong et al., 2022) and to the air-water interfaces (Al Harraq and Bharti, 2022), cycles of wetting and drying (which cause the interfaces to advance and recede repeatedly) play a crucial role. A positive correlation between the number of cycles and the maximum MP depth was reported for various settings (Gao et al., 2021; O'Connor et al., 2019; Ranjan et al., 2023; Zhao et al., 2022b). In addition to interface motion, wetting-drying cycles are associated with soil surface damage, in particular, cracks formed by repeated shrinkage and expansion (Tang et al., 2021; Wan et al., 2019), serving as preferential pathways for MPs. Another mechanism associated with wetting-drying cycles which affects MP transport is mechanical abrasion and weathering of MPs resulting in fragmentation into smaller particles; this was used to explain increased MPs penetration depth with wetting-drying cycles (Ranjan et al., 2023).

5.4. Freeze-thaw cycles

Freeze-thaw cycles can promote MP mobility, by (1) releasing colloids from the soil to which MPs can attach and, therefore, increasing MP mobility; and (2) applying cyclic mechanical pressure on pore walls, creating additional preferential flow pathways (Li et al., 2023; Koutnik et al., 2022b). Contrasting evidence showed that these cycles caused aggregation of MPs, decreasing their mobility (in terms of MP normalized concentration in Alimi et al., 2021). The difference of plastic size was suggested as explanation for these contrasting results (Alimi et al., 2021). The impact of freeze-thaw cycles depends on the type of soil: it was found to increase mobility in loamy sand and sandy soil, with no apparent effect in quartz sand media (in terms of MP normalized concentration) (Li et al., 2023). The authors explained that the loamy sand and sandy soils released more colloids than quartz sand media. However, Koutnik et al. (2022b) found that although a sand and soil mixture (biofilter media) could release more colloids, the freeze-throw cycles increased greater MP mobility (normalized concentration) in sand than that in biofilter media because the pore sizes in sand were larger.

6. Living organisms

Living organisms such as soil animals, plants, and microorganisms, affect MP transport (Huerta Lwanga et al., 2017, 2018; Rillig et al., 2017; Li et al., 2021a; He et al., 2020, 2021), with a larger effect exhibited by larger living species (Ren et al., 2021). It has been shown that soil animals (e.g. earthworms) can promote MP migration by (1) creating macro-pores that serve as preferential MP pathways; and (2) digesting MPs, transporting them further to where MPs are either excreted as defecate or released once the organism dies (Huerta Lwanga et al., 2017; Rillig et al., 2017). Plant roots enhance the penetration depth of MP because their decomposition leaves a macropore for MP transport. MPs in contact with the root tips of plants tend to migrate with the growth of the root system; thus, MP transport will depend on the root orientation, i.e. horizontal vs. vertical (Li et al., 2021a). Rhizosphere secretion can also induce MPs aging and decrease their hydrophobicity, increasing MP mobility (Li et al., 2021a).

Microscopic organisms, i.e. bacteria, affect MP transport by altering both the soil and the MP properties (He et al., 2020, 2021). For example, He et al. (2020) found that the gram-negative strain *E. coli* was shown to decrease the MP mass recovery rate by creating a biofilm which narrows the soil pores as well as increases their surface roughness, as well as by decreasing the repulsive force between MPs and soil particles. Bacteria interact with MPs in a complex manner, by forming MPs-bacteria aggregates (He et al., 2021). Bacteria reduce the Zeta potential of negatively charged MPs (the Zeta potential of MPs-bacteria aggregates is lower than that of only MPs) and increase the hydrodynamic diameter (larger for aggregates than for individual MPs); both effects were shown to decrease the MP mass recovery rate. For positively charged MPs, bacteria will reverse the MP charge (as the overall charge of the aggregates is negative), increasing the MP mass recovery rate in soils with negatively charged grains. In addition, IS influences the effect trend of types of bacteria on MP transport. For example, the reductions of the mass recovery of negatively charged MP by *E. coli* are less than by *B. subtilis* (in both IS = 5 mM and IS = 25 mM). The increase of the mass recovery of positively-charged MP by *B. subtilis* is higher than *E. coli* when the IS was 5 mM; however, this phenomenon was reversed when the IS was 25 mM where *E. coli* increased the zeta potential of MP more negatively than that of *B. subtilis,* likely influencing MP mass recovery (He et al., 2021).

7. Discussion

Microplastic transport is affected by a multitude of physical, chemical, and biological parameters, many of which are intimately coupled. These include properties of: (i) the MPs—size, shape, concentration and type (density, surface roughness and wettability), and the MP zeta potential; (ii) the soil—texture, pore and particle sizes, mineralogy, the zeta potential of soil particle and soil organic matter; (iii) hydrological

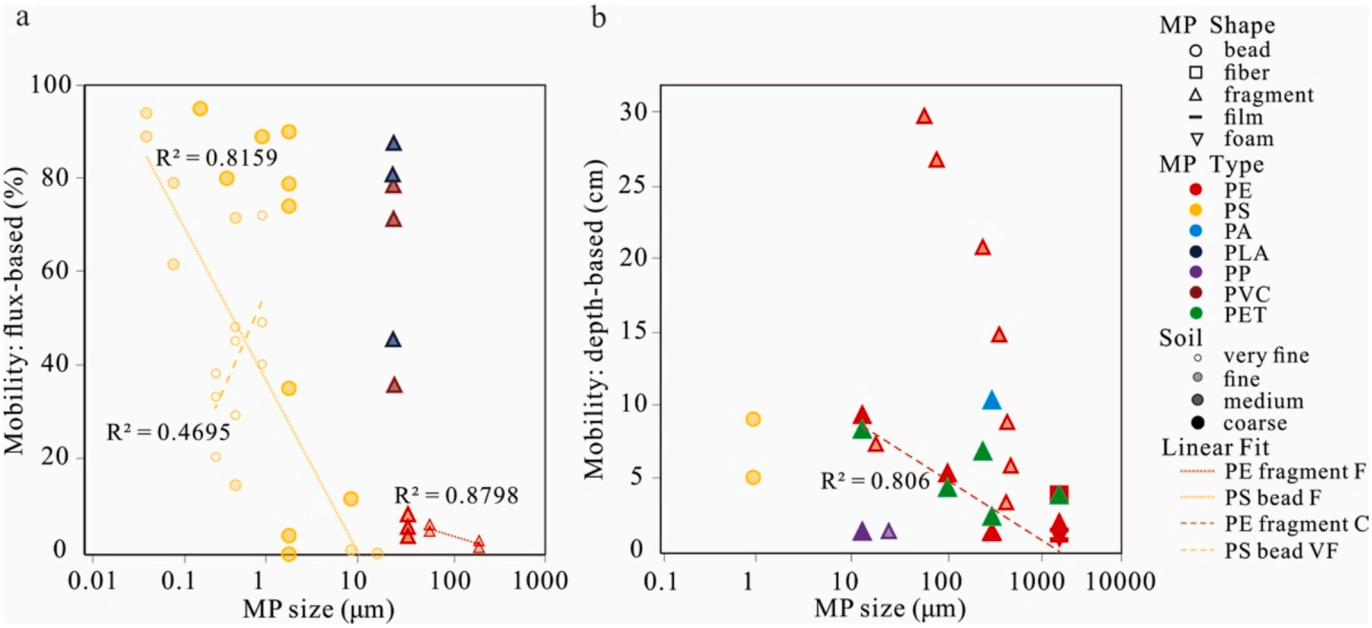


Fig. 4. Complex relationships between microplastic size and their mobility in terms of both (a) transport rate (mass recovery rate, mass balance, total mass recovery), loss ratio, MP concentration, normalized MP concentration and (b) maximum depth (flux- and depth-based, respectively). For data source see Table S6 (SI).

conditions—flow rate, wetting-drying cycles, saturation, and soil solution geochemistry including pH, ionic strength, and chemicals present in soil; and (iv) soil living organisms—plants, animals and microorganisms.

The complex dependence of MP mobility on various processes and properties can be illustrated by relating MP mobility to basic MP properties (type, size, and shape) and soil texture. When all basic parameters are plotted together no clear trend emerges, although some trends can be identified within subsets of the data, i.e. for specific properties (Fig. 4). Given these properties are categorical variables (i.e. with no inherent standalone numerical context), we categorized the data set into 17 unique subsets based on various MP characteristics (Table S7 in SI). However, due to limited amount of data in many of these series, we focus our analysis on a few subsets only. Considering flux-based mobility proxies as the dependent variable, we find for two subsets—(i) PE fragment in fine soil and (ii) PS beads in fine soil—a statistically significant negative correlation between MP size and flux-based mobility (p-values of 0.062 and 0.032, significant at the 10 % and 5 % levels, respectively). In contrast, for PS beads in very fine soil, MP size exhibit a statistically significant positive correlation with flux-based mobility (p-value = 0.046, significant at the 5 % level). For depth-based MP mobility, PE fragments in coarse soil exhibit a statistically significant negative correlation between MP size and depth (p-value = 0.018, significant at the 5 % level, see Figs. S3–5 in SI). Beyond these, no significant correlations were observed for other parameters, suggesting competing effects among these properties. Hence, the effect of some parameters on MP mobility depends on interactions with other parameters.

A formidable source of complexity which challenges our understanding of MP mobility is the interaction of the aforementioned factors to affect mobility, termed here "*co-effects*". By co-effects we refer here to the interference among factors, where the simultaneous changes in two or more parameters alter the competition between various factors and processes. These co-effects are behind the complex and often contradictory findings presented in this review, and the lack of clear trend in MP mobility when plotted vs. a single factor, e.g. the dependence of MP transport on MP size in Fig. 4. While the co-effect of MP sizes and soil pore size is intuitive—larger pore sizes allow larger MPs to pass through, other co-effects of MP size may be less clear. For instance, as MP size

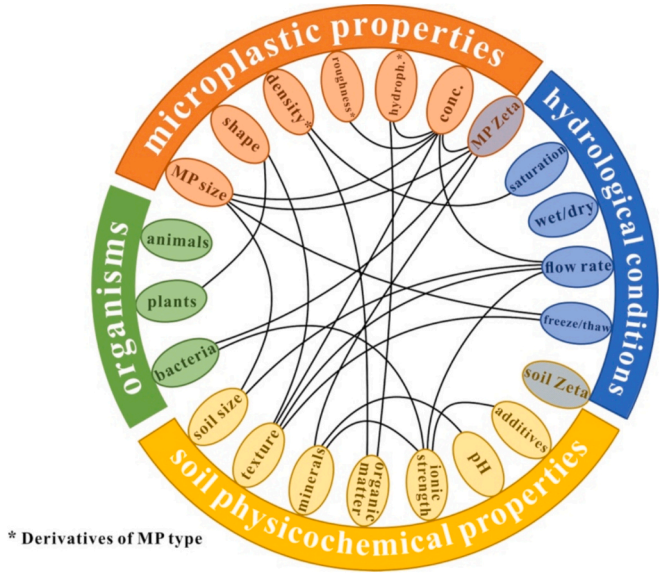


Fig. 5. Factors are shown to affect MP transport (ovals), connected by lines indicating "co-effects" where one factor changes the way the other factor affects MP transport.

decreases below a certain threshold (which in turn could depend on multiple factors such as soil solution chemistry), additional mechanisms which affect transport become more dominant. For MP sizes in the range of 0.1–1 μm, a decrease in MP size decreases their Zeta potential, which enhances MP adsorption on the soil particles, reducing their mobility (reversing the trend for MPs larger than ~1 μm). Aggregation of smaller plastic particles (< 0.1 μm, i.e. NPs) with ionic compounds from the soil solution (a process which by itself is also moderated by the plastics' zeta potential), increases the aggregates' hydrodynamic diameter; this, in turn, can reverse again the mobility trend to be inversely proportional (similar to the inverse proportionality between MP size and mobility in larger MPs, i.e. > 1 μm).

We note that only co-effects evidenced in the literature are presented

in Fig. 5; we expect more co-effects to exist, for example, between the Zeta potentials of the MP and the soil, to be revealed upon further examination. Exploring the interactions among multiple factors and the resulting co-effects (Fig. 5) requires further experimental studies where all parameters besides those investigated are controlled and known. In many cases, multiple parameters that are not the control parameters are unknown or vary together with other parameters, challenging conclusive evidence on the specific effect of the actual control parameter of interest.

Another challenge in our understanding of MP mobility in soils is the lack of standard metrics and standard methods of analysis quantifying it. Currently, published studies vary in the application of sampling methods (e.g. collecting effluent samples from the column transport experiments, extracting MPs from soil or sediment samples), in sample preparation methods for MP analysis (e.g. sonication, magnetic stirring, wet sieving, density separation using different chemicals, digestion of organic matter), and in methods for identification and quantification of MPs (e.g. Fourier transform infrared spectroscopy, UV–Vis spectrophotometry, fluorescence spectrophotometry, stereomicroscopy, scanning and transmission electron microscopy) (Table S5). Similarly, there are multiple metrics which are defined differently for MP mobility (Table 1). While some of these metrics are correlated (meaning that higher values of these metrics indicate greater MP mobility), they do not represent identical phenomena nor carry the same environmental implications, and thus cannot be used interchangeably. For example, a scenario where only a few MPs reach very deep would be characterized by a low transport rate and loss ratio, however a large maximum penetration depth (Zhang et al., 2022b). These differences in methods of analysis and in metrics not only hinder the cross-comparison of published studies and the development of a unified understanding of MP behavior in soils, but also limit the applicability of findings to broader environmental contexts. Addressing this challenge thus requires the development and adoption of standardized protocols and metrics that enable consistent and reproducible evaluations of MP transport in soil.

8. Conclusion and future perspectives

In conclusion, our ability to predict MP transport in soils is hindered by several knowledge gaps, owing to the complex underlying physics, chemistry, and biology. This scientific challenge makes MP transport a vibrant, exciting topic that is becoming intensely explored by various disciplines. We point to the need for further fundamental understanding of the transport mechanisms controlling how hydrophobic MP particles move in heterogenous porous materials; some of this knowledge can only be addressed by first studying idealized systems, followed by more applied studies which incorporate the complexity of natural soils and the various environmental conditions. Specifically, we flag the following research needs:

(i) Gaining fundamental insights into the physics of the MP transport process, in particular for very small MPs where colloidal forces may become important, through the use of soil analogs such as glass beads and micromodels; these insights should be then tested against experiments using natural soils, in particular clay-rich, where the complex geochemistry and pore size distribution may govern MP transport;
(ii) Extending the range of MP sizes for different MP types; currently all studies on very small MPs (< 20 μm) are performed with polystyrene particles only;
(iii) Understanding the transport of MPs when other particles adsorb on them, and the potential impact of changes in the surface properties of the aggregate (vs. the MP alone); this is also highly-relevant for transport of heavy metals, pathogens, and antibiotics by MPs acting as vectors;
(iv) Examining the role of preferential pathways, which prevail in hydrophobic soils, and the role played by the hydrophobic MPs—known to induce such pathways, on MP transport;
(v) Exposing the impact of the temporal evolution of MP properties as they age and degrade with changing environmental conditions;
(vi) Identifying additional interactions among multiple factors and co-effects.

We note that some of the controlling factors are less studied than others, probably due to technical challenges rather than their lesser importance; for example, the role of MP shape and of MP composites (as many plastics are not made of a single polymer). Another, more technical challenge is the use of different MP mobility metrics, e.g. maximum depth reached within a sample vs. concentration traversing the entire sample, which refer to disparate aspects of MP mobility that cannot be compared quantitatively, and of different environmental implications. The development of standardized analytical methods for the identification and quantification of microplastics in soil and their comparison in benchmark studies are essential to facilitate meaningful comparisons, identify consistent trends, and enhance the reliability and impact of quantitative reviews across research studies. The broad, interdisciplinary community studying MP transport would benefit from the research incorporating the above aspects in a systematic and synergetic manner. These scientific advancements can also improve our ability to monitor and mitigate MP pollution, as well as develop less-polluting materials, and improve policies for waste management, treatment, and mitigation.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix A. Supplementary data

Supplementary data to this article can be found online at https://doi.org/10.1016/j.earscirev.2025.105108.

Data availability

Data provided as SI.

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